OUTCOME	CLINICAL ONLY		PET ONLY		COMBINED	
	VALIDATION	TEST	VALIDATION	TEST	VALIDATION	TEST
ROC-AUC	0.9420	0.9404	0.9004	0.5491	0.974638	0.856683
ACCURACY	0.9351	0.8974	0.8961	0.8718	0.922078	0.935897
BALANCED ACCURACY	0.7428	0.7488	0.5553	0.5411	0.680254	0.770531
SENSITIVITY	0.9855	0.9420	0.9855	0.9710	0.985507	0.985507
SPECIFICTY	0.5000	0.5556	0.1250	0.1111	0.375000	0.555556
PRECISION	0.9444	0.9420	0.9067	0.8933	0.931507	0.944444
F1 – SCORE	0.9645	0.9420	0.9444	0.9306	0.957746	0.964539
PR- AUC	0.9933	0.992	0.9880	0.9061	0.997189	0.975493
BRIER SCORE	0.0626	0.0784	0.0850	0.1202	0.057970	0.065225

CONFUSION MATRIX	VALIDATION	TEST
CLINICAL	Validation Confusion Matrix Predicted HPV 0 Predicted HPV 1 Actual HPV 0 4 4 Actual HPV 1 1 68	Test Confusion Matrix Predicted HPV 0 Predicted HPV 1 Actual HPV 0 5 4 Actual HPV 1 4 65
PET	Validation Confusion Matrix Predicted HPV 0 Predicted HPV 1 Actual HPV 0 1 7 Actual HPV 1 1 68	Test Confusion Matrix Predicted HPV 0 Predicted HPV 1 Actual HPV 0 1 8 Actual HPV 1 2 67
COMBINED	Confusion Matrix (rows=true, columns=predicted) Predicted HPV negative Predicted HPV positive Actual HPV negative 3 5 Actual HPV positive 1 68	Confusion Matrix (rows=true, columns=predicted) Predicted HPV negative Predicted HPV positive Actual HPV negative 5 4 Actual HPV positive 1 68

CALIBRATION ANALYSIS

CLINICAL :

Validation Calibration Table

bin_range	sample_count	mean_predicted_probability	observed_hpv_positive_rate	absolute_calibration_gap
[0.0, 0.1)	3	0.0103	0.0000	0.0103
[0.1, 0.2)	1	0.1338	1.0000	0.8662
[0.2, 0.3)	0	NaN	NaN	NaN
[0.3, 0.4)	1	0.3739	0.0000	0.3739
[0.4, 0.5)	0	NaN	NaN	NaN
[0.5, 0.6)	1	0.5926	1.0000	0.4074
[0.6, 0.7)	0	NaN	NaN	NaN
[0.7, 0.8)	2	0.7662	1.0000	0.2338
[0.8, 0.9)	2	0.8874	0.5000	0.3874
[0.9, 1.0]	67	0.9934	0.9552	0.0382

Test Calibration Table

bin_range	sample_count	mean_predicted_probability	observed_hpv_positive_rate	absolute_calibration_gap
[0.0, 0.1)	4	0.0176	0.0000	0.0176
[0.1, 0.2)	4	0.1350	0.7500	0.6150
[0.2, 0.3)	1	0.2288	1.0000	0.7712
[0.3, 0.4)	0	NaN	NaN	NaN
[0.4, 0.5)	0	NaN	NaN	NaN
[0.5, 0.6)	0	NaN	NaN	NaN
[0.6, 0.7)	1	0.6495	0.0000	0.6495
[0.7, 0.8)	0	NaN	NaN	NaN
[0.8, 0.9)	1	0.8777	1.0000	0.1223
[0.9, 1.0]	67	0.9939	0.9552	0.0387

PET :

Validation Calibration Table

bin_range	sample_count	mean_predicted_probability	observed_hpv_positive_rate	absolute_calibration_gap
[0.0, 0.1)	0	NaN	NaN	NaN
[0.1, 0.2)	0	NaN	NaN	NaN
[0.2, 0.3)	1	0.2301	1.0000	0.7699
[0.3, 0.4)	0	NaN	NaN	NaN
[0.4, 0.5)	1	0.4738	0.0000	0.4738
[0.5, 0.6)	2	0.5659	0.5000	0.0659
[0.6, 0.7)	1	0.6495	1.0000	0.3505
[0.7, 0.8)	1	0.7481	0.0000	0.7481
[0.8, 0.9)	4	0.8891	0.5000	0.3891
[0.9, 1.0]	67	0.9949	0.9552	0.0397

Test Calibration Table

bin_range	sample_count	mean_predicted_probability	observed_hpv_positive_rate	absolute_calibration_gap
[0.0, 0.1)	1	0.0273	1.0000	0.9727
[0.1, 0.2)	1	0.1791	0.0000	0.1791
[0.2, 0.3)	0	NaN	NaN	NaN
[0.3, 0.4)	0	NaN	NaN	NaN
[0.4, 0.5)	1	0.4323	1.0000	0.5677
[0.5, 0.6)	0	NaN	NaN	NaN
[0.6, 0.7)	2	0.6647	1.0000	0.3353
[0.7, 0.8)	1	0.7771	1.0000	0.2229
[0.8, 0.9)	2	0.8987	0.5000	0.3987
[0.9, 1.0]	70	0.9965	0.9000	0.0965

COMBINED

Calibration Table

Bin range	Sample count	Mean predicted probability	Observed HPV positive rate	Absolute calibration gap
[0.0, 0.1)	2	0.029654	0.000000	0.029654
[0.1, 0.2)	0	NaN	NaN	NaN
[0.2, 0.3)	1	0.292989	1.000000	0.707011
[0.3, 0.4)	1	0.311743	0.000000	0.311743
[0.4, 0.5)	0	NaN	NaN	NaN
[0.5, 0.6)	1	0.575881	0.000000	0.575881
[0.6, 0.7)	1	0.693239	1.000000	0.306761
[0.7, 0.8)	0	NaN	NaN	NaN
[0.8, 0.9)	1	0.802041	0.000000	0.802041
[0.9, 1.0]	70	0.996509	0.957143	0.039366

Calibration Table

Bin range	Sample count	Mean predicted probability	Observed HPV positive rate	Absolute calibration gap
[0.0, 0.1)	2	0.014236	0.000000	0.014236
[0.1, 0.2)	2	0.128265	0.500000	0.371735
[0.2, 0.3)	1	0.208175	0.000000	0.208175
[0.3, 0.4)	1	0.332287	0.000000	0.332287
[0.4, 0.5)	0	NaN	NaN	NaN
[0.5, 0.6)	1	0.580324	1.000000	0.419676
[0.6, 0.7)	0	NaN	NaN	NaN
[0.7, 0.8)	0	NaN	NaN	NaN
[0.8, 0.9)	0	NaN	NaN	NaN
[0.9, 1.0]	71	0.995134	0.943662	0.051472

CLINICAL :

Expected Calibration Error	0.0712	0.0854
TN	4.0000	5.0000
FP	4.0000	4.0000
FN	1.0000	4.0000
TP	68.0000	65.0000

PET :

Expected Calibration Error	0.0869	0.1304
TN	1.0000	1.0000
FP	7.0000	8.0000
FN	1.0000	2.0000
TP	68.0000	67.0000

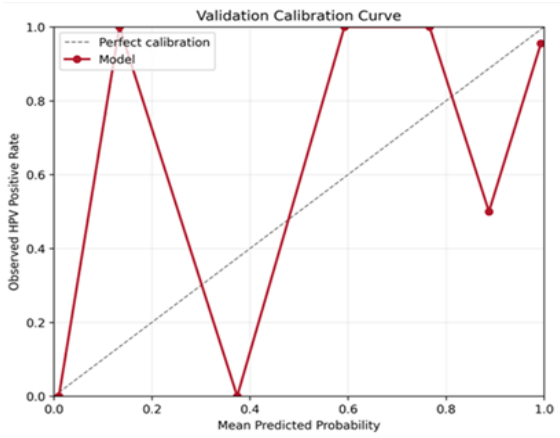
COMBINED :

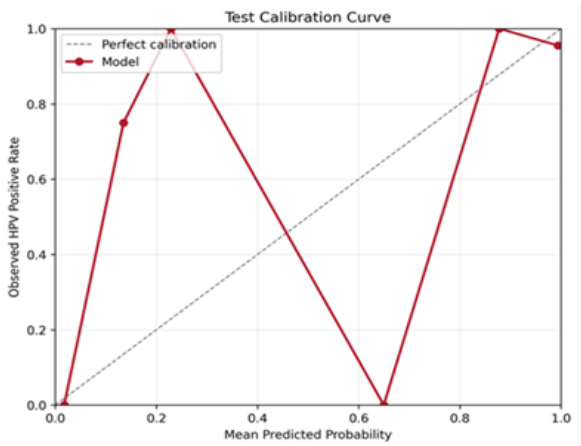
Expected Calibration Error (ECE): 0.071667

Expected Calibration Error (ECE): 0.069059

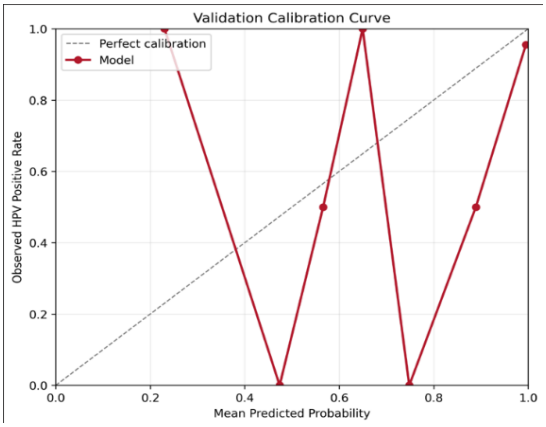
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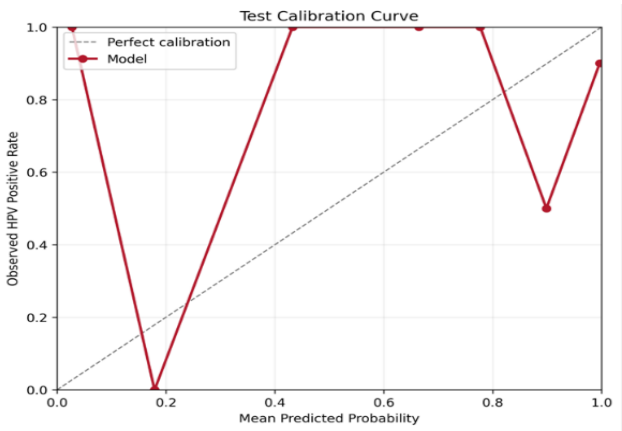
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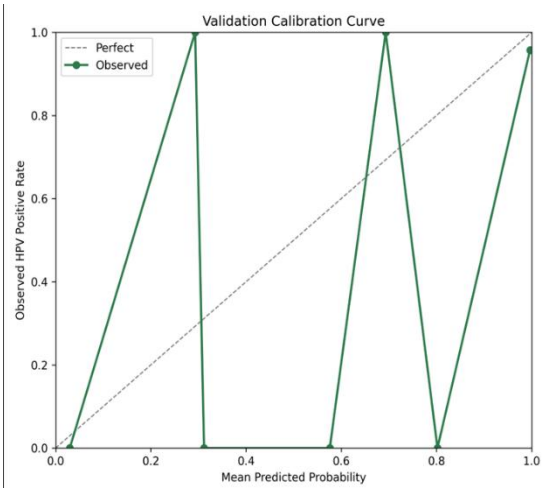


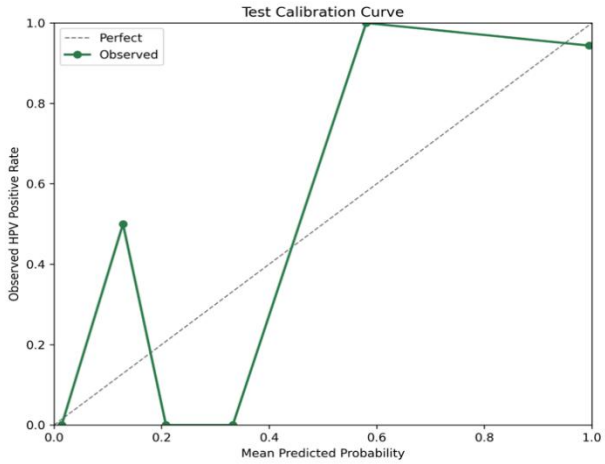
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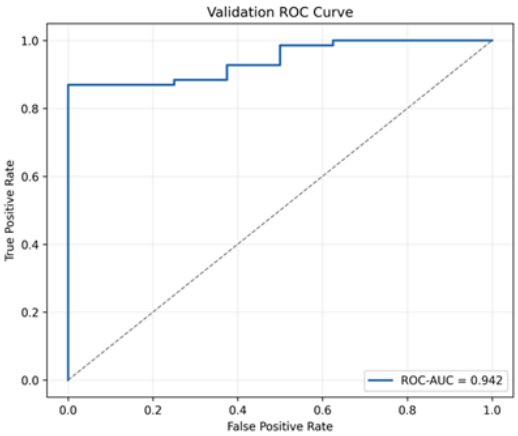


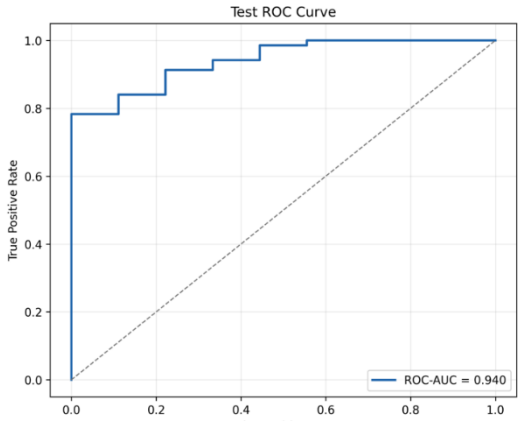
COMBINED :



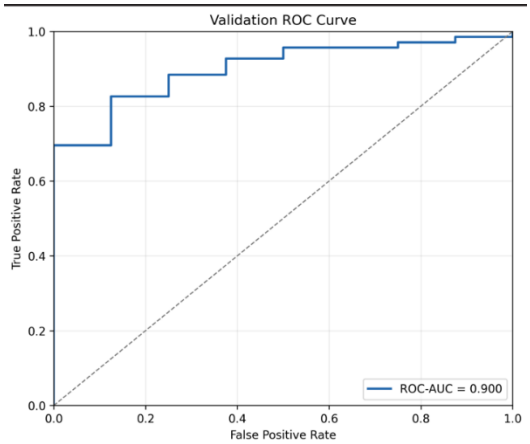


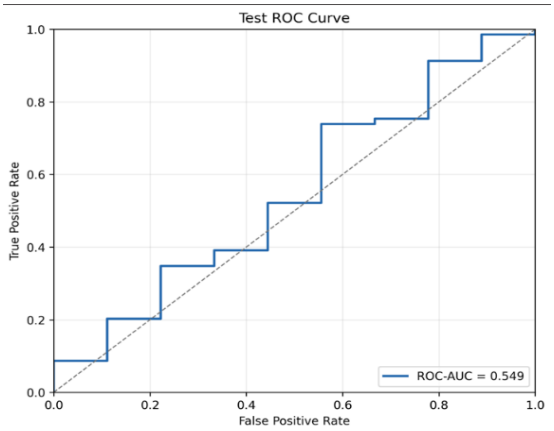
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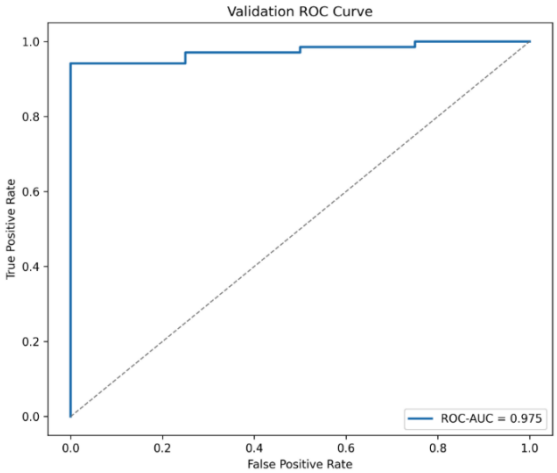


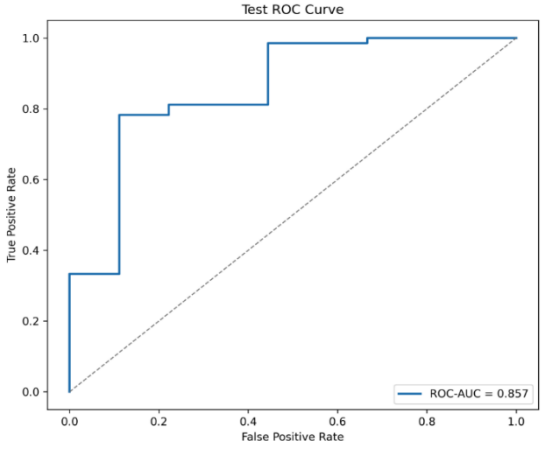
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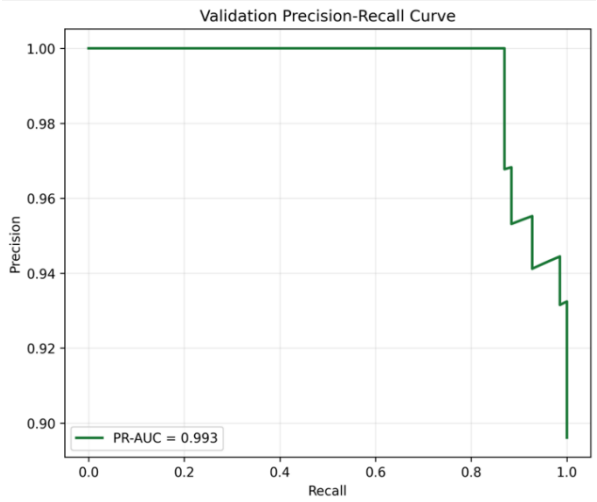


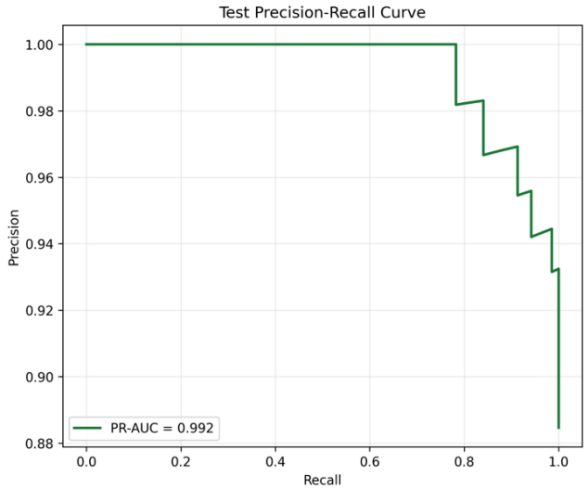
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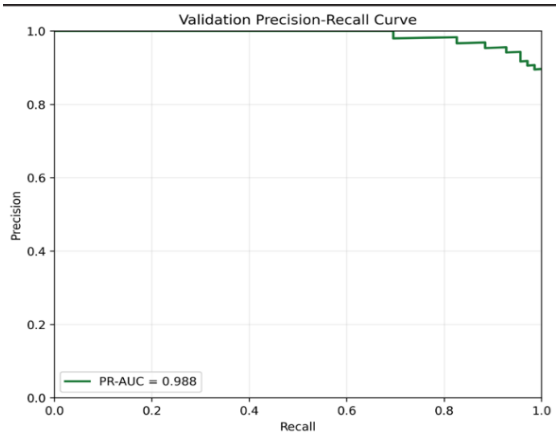


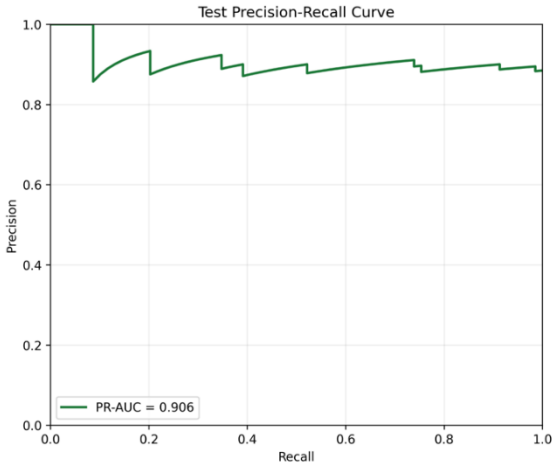
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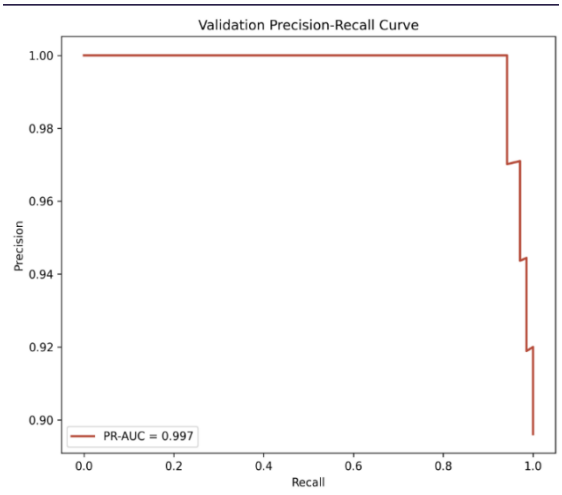


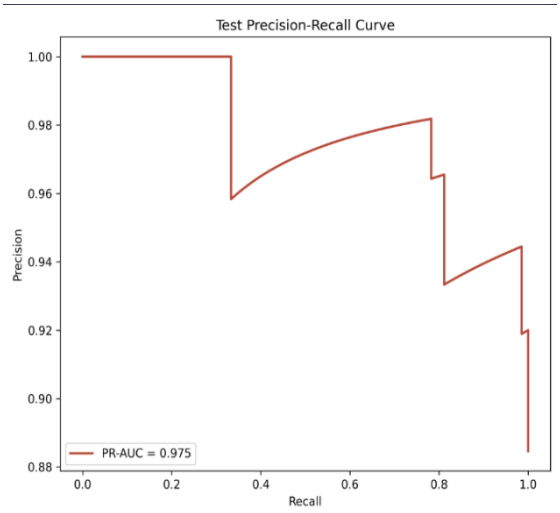
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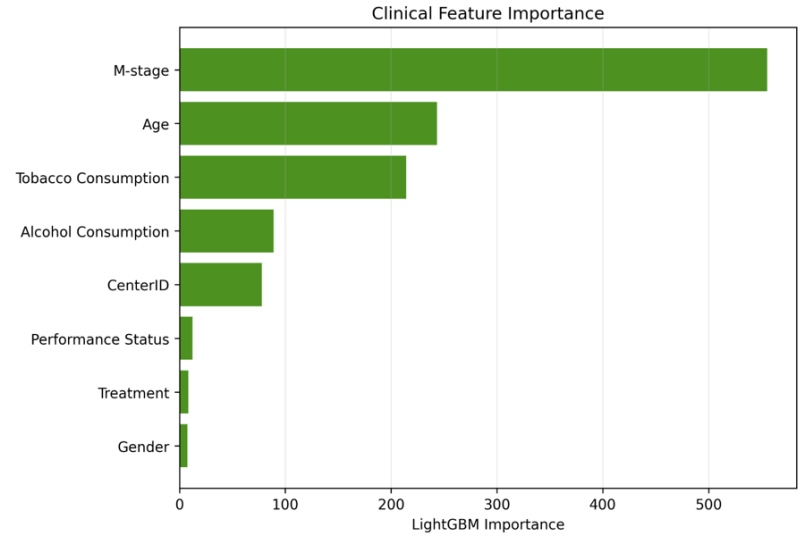


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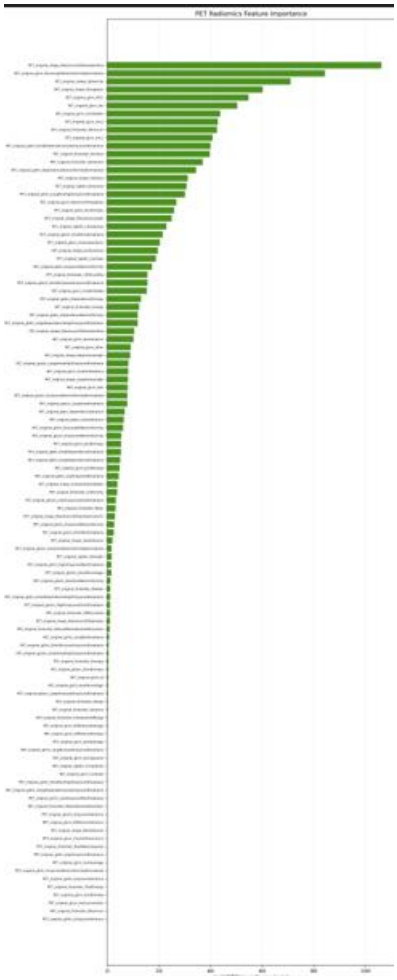




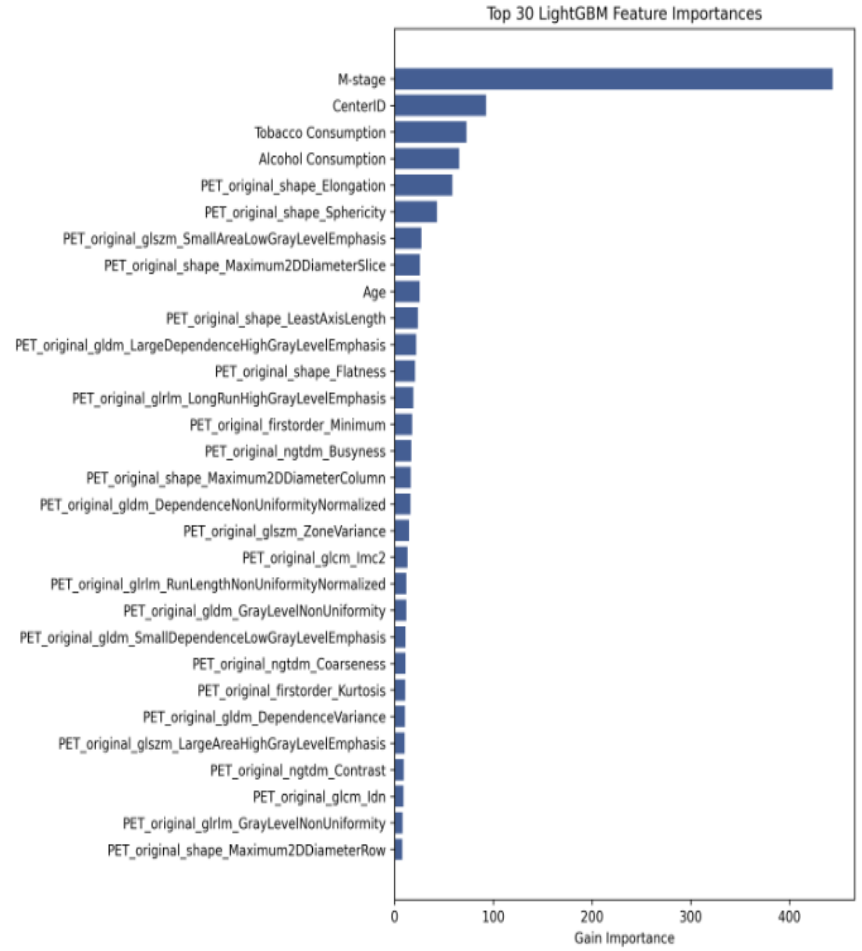
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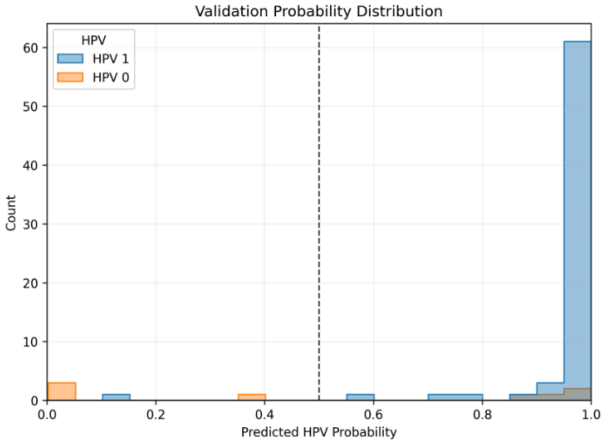
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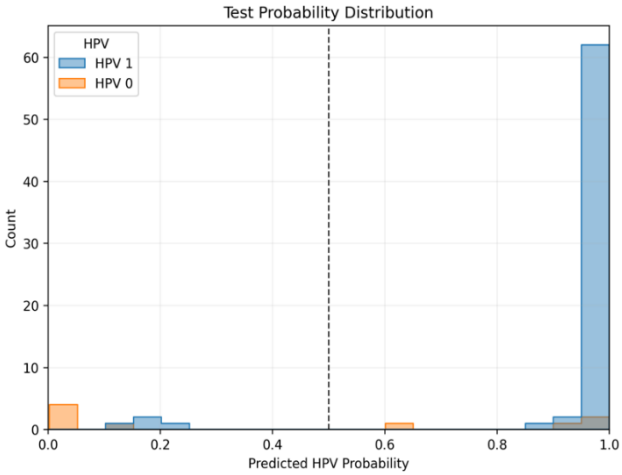


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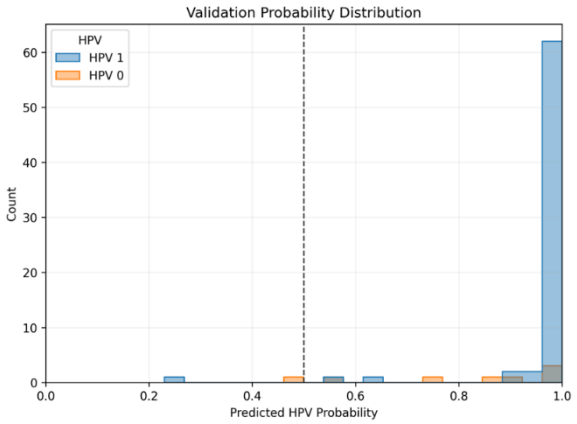


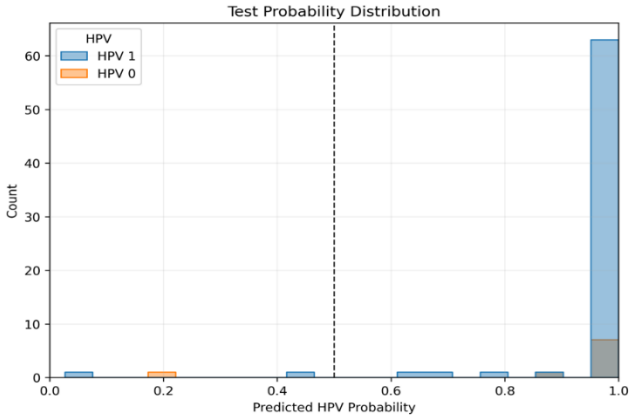
CLINICALP



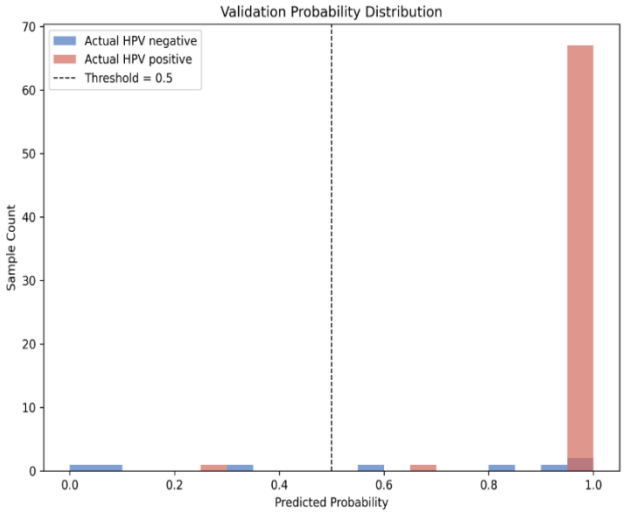


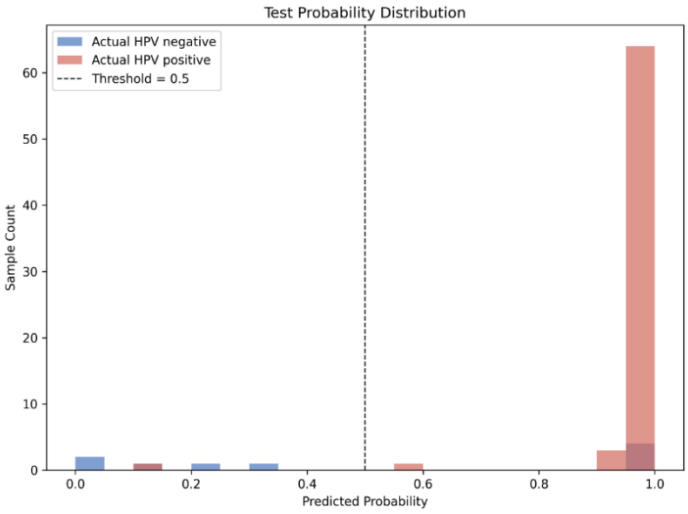
PET:



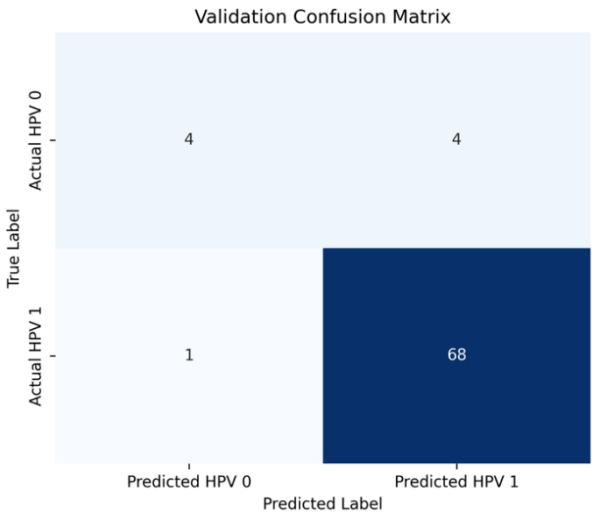


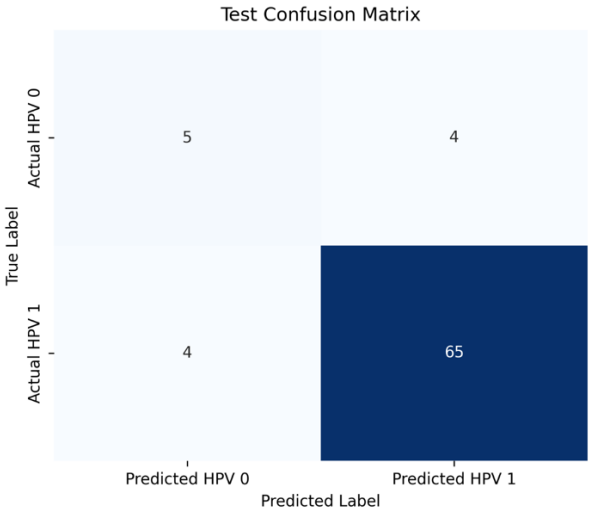
COMBINED:



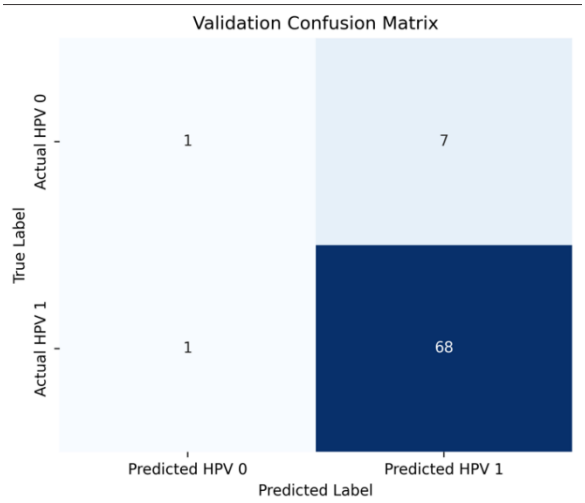


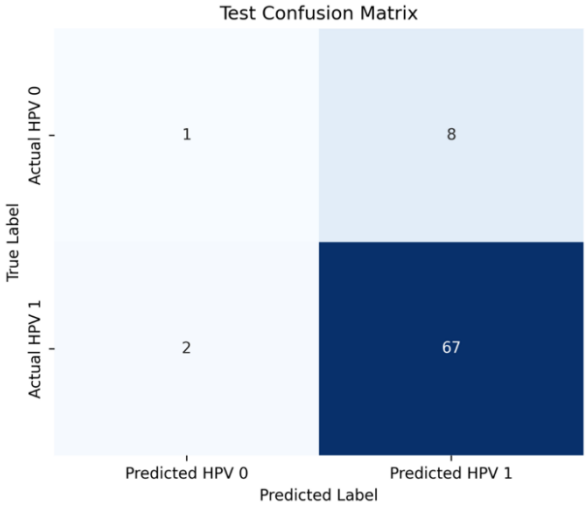
CLINICAL:



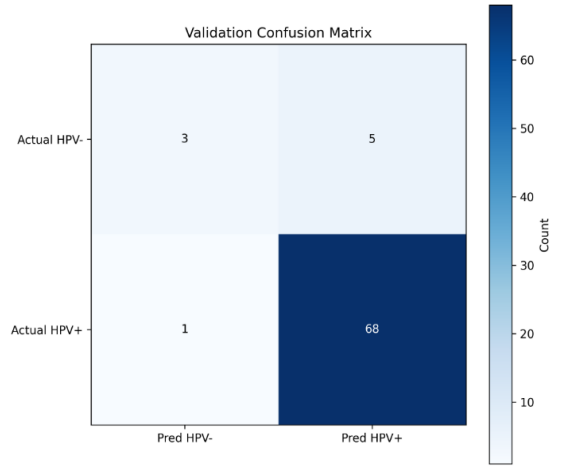


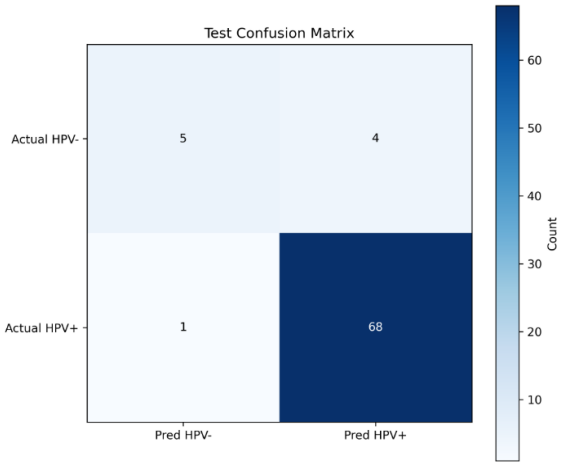
PET:





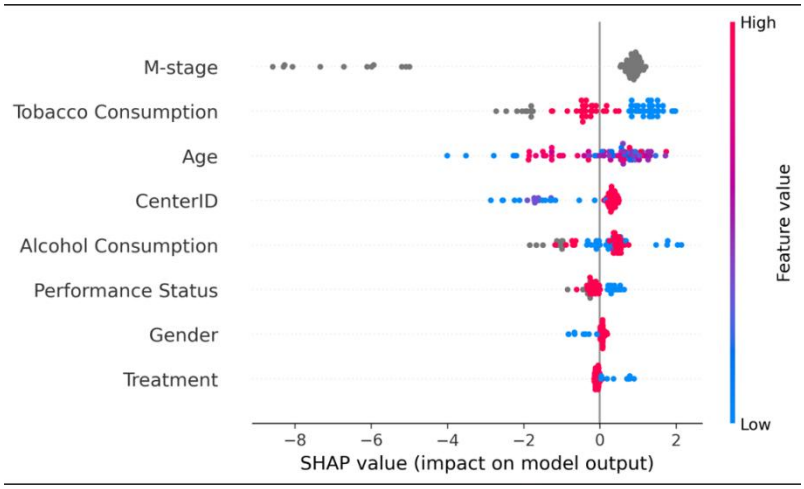
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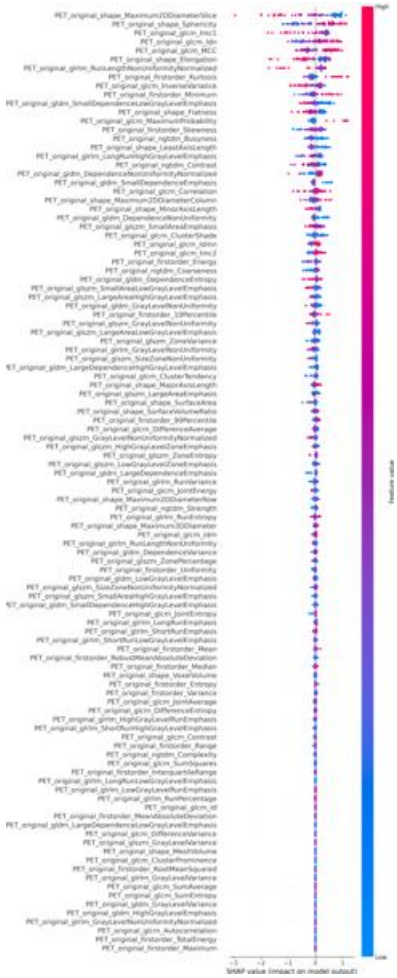


SHAP SUMMARY

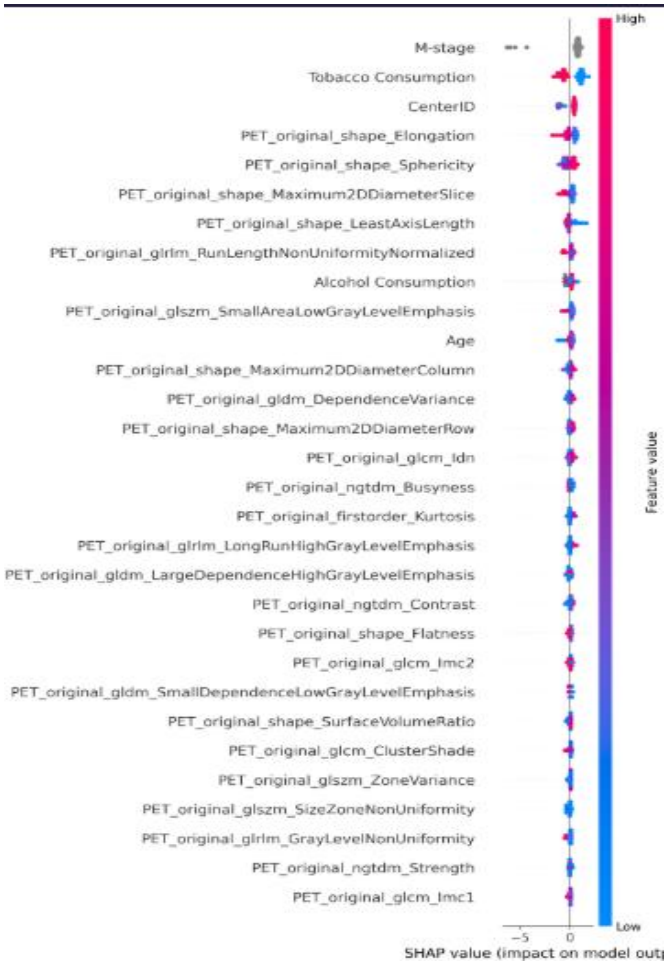
CLINICAL:



PET:

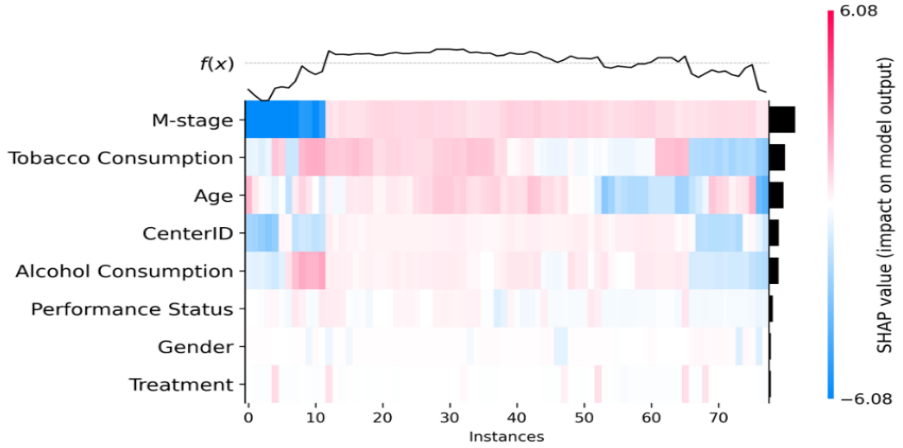


COMBINED:

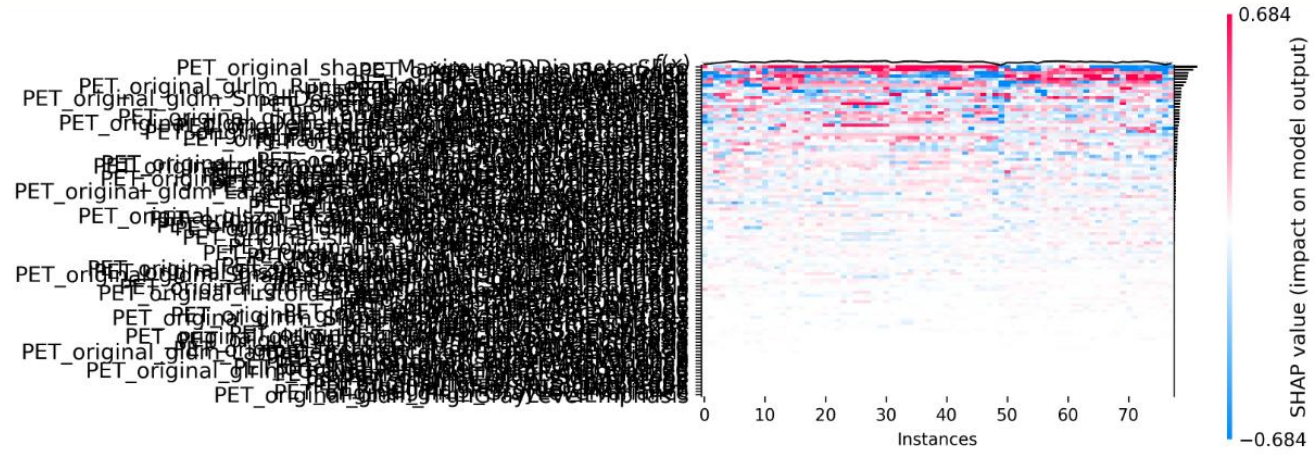


SHAP HEATMAP

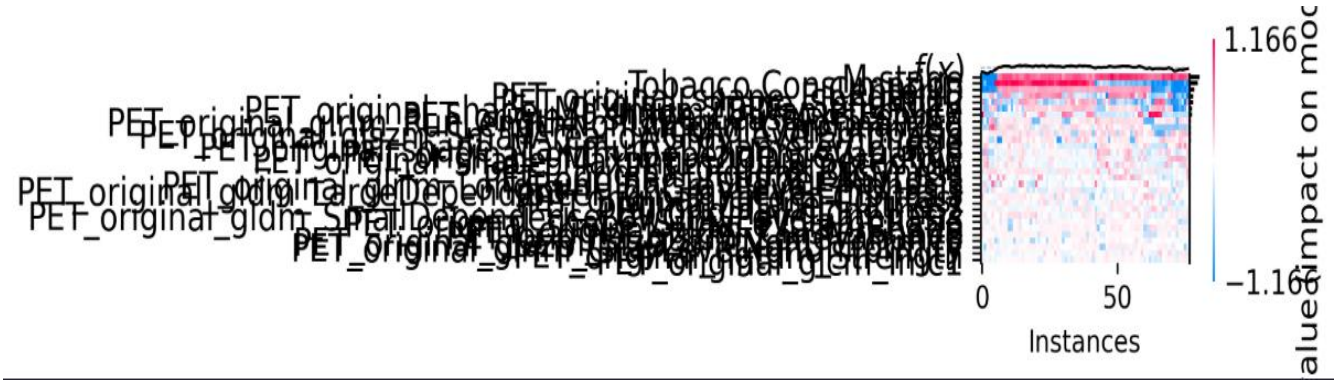
CLINICAL:



PET:

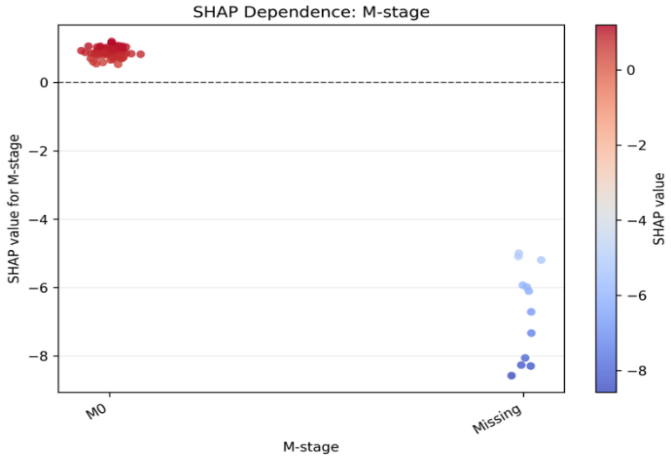


COMBINED:

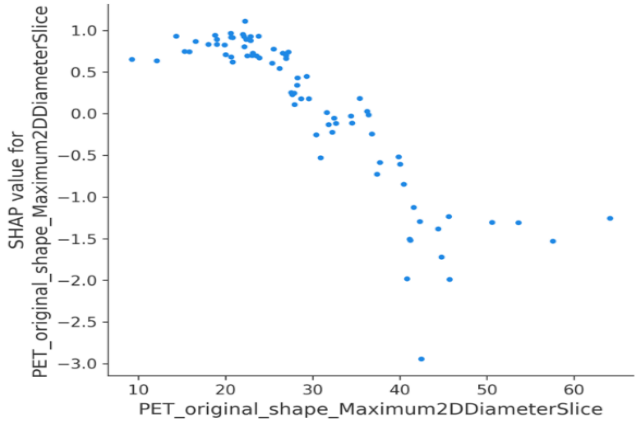


SHAP DEPENDENCE

CLINICAL:



PET:



COMBINED:

